

SAMONWITA SARKER

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Education

Chittagong University of Engineering and Technology (CUET)

2023 – 2027 (Expected)

B.Sc. in Computer Science and Engineering; CGPA: 3.51/4.00 (latest term: 3.85)

Chattogram, Bangladesh

Experience

BdREN Innovation (Cortex)

Sep 2026

Team Lead, Crawler Team (Industrial Attachment) | [GitHub](#)

Bangladesh

- Led the five-person crawler team for Cortex, which crawls a website into a vector database and answers questions with citations; split the work into owned portions and took it through review.
- Built the CUET scraper in **Python** (JSON API first, headless-browser fallback, 26-check verification), capturing **735 documents**, **382 faculty profiles**, and **1,106 linked files** with source provenance.

Fleet AI, Inc.

May 2026 – Aug 2026

AI Evaluation Contractor

Remote

- Authored **70+ long-horizon agentic AI evaluation tasks** across simulated enterprise environments.
- QA-reviewed peers' tasks for rubric coverage, action counts, ambiguity, and project-rule compliance.

PIERRA

2025

Freelance Web Developer | [Website](#)

Montreal (Remote)

- Shipped a bilingual (EN/FR) site for an exterior design firm, brief to launch, in **Next.js** and **Tailwind CSS**.
- Kept all copy in one locale-keyed layer so both languages stay in step; built the gallery and consultation form.

Projects

Greenlight: Robot Arm Simulator | [Next.js](#), [Three.js](#), [TypeScript](#), [Zustand](#)

2026 | [Live](#) | [GitHub](#)

- Led development of an **IUT Techathon 2026 finalist**: a 3D robot arm driven six ways (joystick, keyboard, voice, a Gemini voice agent, PIN entry, natural language) through one motion pipeline.
- Wrote the **IK planner**, the **safety gate** that validates every move, the executor, and the input paths.

Sixpence: Personal Finance Ledger | [PHP 8](#), [PostgreSQL](#)

2026 | [Live](#) | [GitHub](#)

- Built and deployed a finance app with **no framework and no third-party dependencies**, writing the PDF export engine and SMTP client from scratch; budgets, recurring bills, savings goals, and reports in 12 currencies.
- Secured with prepared statements, CSRF tokens, logout, and a strict CSP; **68 tests** run in CI on PHP 8.1–8.4.

DimSum: LLM-Assisted Energy Optimizer | [Bun](#), [TypeScript](#), [Gemini API](#)

2026 | [Live](#) | [GitHub](#)

- Built the Gemini stage turning plain-English operator notes into structured directives for a 24-hour campus energy plan, and a **deterministic guardrail layer** that repairs or discards unsafe model output.
- Cut median model latency from **5.7s to 2.3s**; all ten published scenarios match the organisers' reference cost.

Chrome Extensions (Solo): **RIPHours** & **TabSaver** | [JavaScript](#), [Chrome APIs](#)

2025 | [RIPHours](#) | [TabSaver](#)

- RIPHours (300+ installs)**: privacy-first time tracking with per-site limits and zero network requests.
- TabSaver (400+ installs)**: restores whole tab groups in one click, freeing memory without losing context.

Research & Publications

- A Machine Learning and Explainable AI-Based Multiclass Police Fraud Prediction Scheme with SHAP Based Interpretability.** M. Chowdhury, J. Islam, Md. A. I. Semon, **S. Sarker**, M. M. Barua, A. Akter. *ICECTE 2026*, IEEE. [IEEE Xplore](#)
- 7th place, DravidianLangTech @ ACL 2026 Shared Task (Telugu Prompt-Style Recovery).** First-author system paper on nine-way Telugu style classification. [Paper](#) | [Code](#)

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, Java, C, PHP, SQL | **Tools:** Git, Docker, Linux, LaTeX

Web: React, Next.js, Node.js, Express.js, PostgreSQL, Firebase, Tailwind CSS, Three.js

AI/ML: PyTorch, scikit-learn, Hugging Face Transformers, NLP, multimodal deep learning, SHAP, Gemini API

Leadership & Competitions

- Leadership:** President, *CUET MUN Club* | Vice President (Organizing), *CUET Computer Club* | General Secretary, *IEEE CS CUET Development Wing* | Joint General Secretary, *CUET Debating Society*
- Competitions:** Finalist at IUT Techathon 2026, SciBlitz 2.0 AI Hackathon 2026, Infinity AI BuildFest 2026, PoliMemeDecode Datathon (CUET CSE Fest 2025), and MicroOps Hackathon 2025.